Shahjalal University of Science and Technology
Department of Computer Science and Engineering
4th Year 1st Semester Final Examination (Session: 2020-21)
Course Code: CSE 421, **Credits:** 2 **Course Title:** Software Project Management
Time: 2 hrs **Total Marks:** 50 **Group A**

[Answer all the questions]

1) ***Answer any five(5) questions from the following:***

a) Define Internal Stakeholders in the context of Software Project Management (SPM) with suitable examples. 1

b) Define Generic Software and illustrate with appropriate examples. 1

c) Explain the term "Code-Like-Hell Programming" as a classical mistake in SPM. 1

d) Explain the term "Short-Changed Upstream Activities" as a classical mistake in SPM. 1

e) Define project scope in Software Project Management with an example. 1

f) Explain the PDCA Loop with an example in software development. 1

g) Write the full form of SMART in the context of identifying project objectives. 1

2) ***Answer any four (4) questions from the following:***

a) Define the term *Software Crisis*. What were the underlying reasons behind it? Give one suitable example. 2.5

b) Project Management consists of five process groups: Initiating, Planning, Executing, Monitoring and Controlling and Closing. Explain each with an example. 2.5

c) What is a *CI/CD Pipeline*? Explain how it works in modern software development. 2.5

d) Distinguish between *risk* and *opportunity* using both real-life and software engineering examples. 2.5

e) Explain the concepts of *Person-Month*, *Brooks' Law*, and *Cost Drivers* in relation to the *COCOMO model*. 2.5

f) Suppose you are planning to develop a video platforming system like YouTube. Describe the entire development process of this system in terms of *Software Project Management*. 2.5

3) ***Answer any two (2) questions from the following:***

a) What is the W5HH principle in Software Project Management? Explain it with suitable examples. 5

b) Industry "Messi" is developing a payroll system for a small company. This is considered a small and simple project.
Industry "CR7" is developing a flight control system for the SU-57 air fighter plane, which is a very complex, real-time, and hardware-dependent project.Using the **Basic COCOMO model** and the given constant table, analyze and compare the following for both projects (assume KLOC = 40): 5

- Effort Estimation (Person-Months)
- Development Time (Schedule in Months)
- Number of People required

Constants:

Mode	a	b	c	d
Organic	2	1	2	1
Semi-detached	3	2	2	1
Embedded	4	3	2	1

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c) At the start of the COVID-19 pandemic, Slack's server could not handle the surge in remote work traffic. Business worldwide faced disruptions when Slack went down multiple times. The issue highlighted lack of preparation for demand spikes. 5
Which type of risk is this and why? Explain how can you minimize this type of risk.

Group B

Answer any five(5) questions from the following:

4)

a) Define Continuous Process Improvement(CPI) in the context of software projects. 1

b) Define MoSCoW Prioritization technique with examples. 1

c) Write any two reasons why Risk Management is important in software projects. 1

d) Explain the Blended Approach for risk management. 1

e) State the purpose of Jenkins Software in software development. 1

f) Define Software Process Improvement (SPI) and explain its purpose. 1

g) Define Risk Mitigation in software project management. 1

5) ***Answer any four (4) questions from the following:***

a) Write down the key clauses of ISO 9001. 2.5

b) Why is Work Breakdown Structure (WBS) important when defining project scope? 2.5

c) Describe the role of a Project Manager in defining project planning and project scope with example. 2.5

d) Differentiate between proactive and reactive strategies for risk management with suitable examples. 2.5

e) How is Software Project Management differ from traditional Project Management? 2.5

f) Discuss the major factors that cause software project failure. 2.5

6) ***Answer any two (2) questions from the following:***

a) Analyze the different levels of a Work Breakdown Structure (WBS) for developing an E-Commerce website. Explain the key components included at each level. 5

b) A software firm has standardized processes. Every team follows the same SDLC phases (requirements, design, coding, testing, deployment) with documented checklists. Training sessions are given to all new employees. 5
At which CMM level is this company? Explain all the maturity levels from this identified maturity level to the highest maturity level illustrating the key characteristics of maturity levels

c) A company worries about losing customer data. It reduces likelihood by using reliable databases, reduces impact with daily backups, transfers risk with cyber-insurance, and accepts small risks of minor data corruption. - 5

Explain four Risk Mitigation Strategies with example according to mentioned scenario.

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